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Savitribai Phule Pune University

Information Technology - Third Year

Computer Network and Security - Unit II

WIRELESS STANDARDS

Wireless LANs: Fundamentals of WLAN, Design goals, Characteristics, Network Architecture, IEEE 802.11: components in IEEE 802.11 network, Physical Layer, MAC Sub Layers : DCF, PCF, Hidden and exposed station problem, Frame format, Addressing Mechanism, IEEE 802.15.1 Bluetooth: Architecture, Layers, operational states, IEEE 802.16 WiMax: Services, Architecture, Layers, comparisonbetween Bluetooth, IEEE 802.11 and IEEE 802.16

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Q1. Explain the Basic Architecture of WLAN and Discuss Various Components in it.

Definition

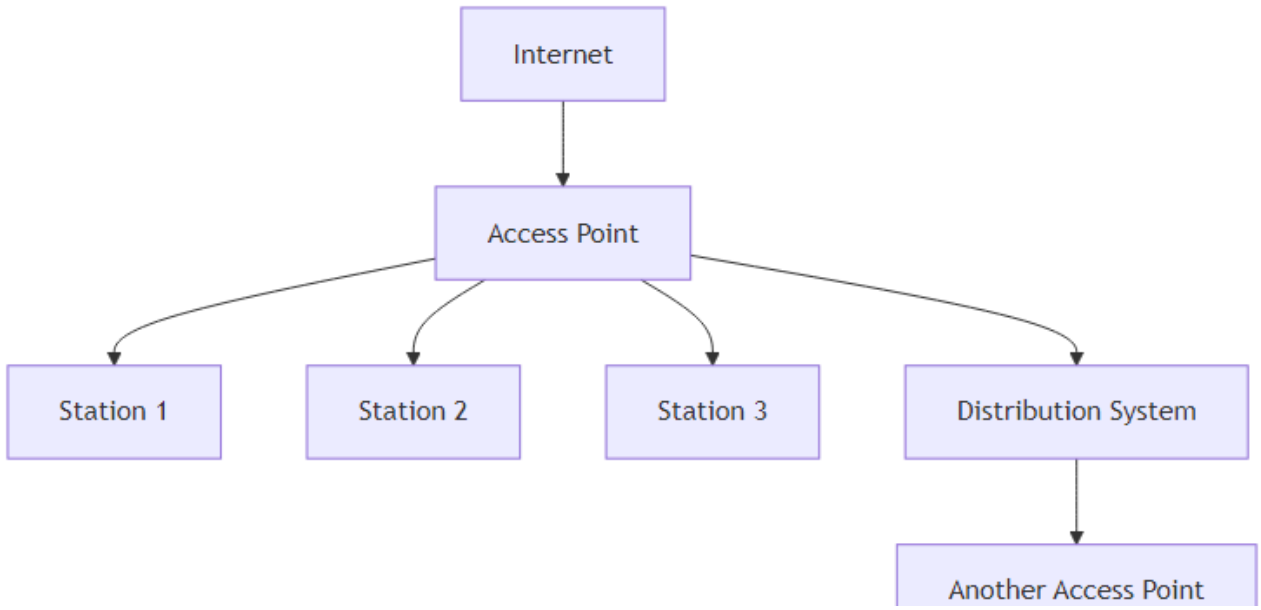
WLAN (Wireless Local Area Network) is a wireless network that connects devices using radio waves instead of cables.

It is based on IEEE 802.11 standards.

It allows devices like laptops, mobiles, and printers to connect within a limited area such as home, office, or campus.

Basic Architecture of WLAN

Diagram



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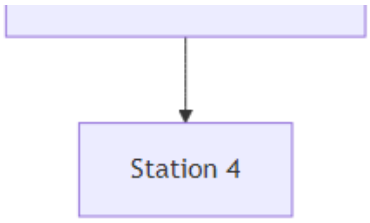

Caption: Basic WLAN Architecture with Access Points and Stations.

Components of WLAN

1. Station (STA)

- Any wireless device
- Laptop, mobile, printer
- Connects to Access Point

2. Access Point (AP)

- Central device in WLAN
- Connects wireless devices
- Acts as bridge between wireless and wired network

3. Basic Service Set (BSS)

- Group of stations connected to one AP
- Identified by BSSID

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4. Extended Service Set (ESS)

- Multiple BSS connected together
- Used in large areas like campuses

5. Distribution System (DS)

- Connects multiple access points
- Usually wired backbone

6. Wireless Medium

- Uses radio frequency (2.4 GHz / 5 GHz)

Types of WLAN Architecture

1. Infrastructure Mode → Uses Access Point
2. Ad-hoc Mode → Direct device-to-device communication

Conclusion

WLAN architecture consists of stations, access points, and distribution systems. It enables wireless communication within limited area.

Example

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Wi-Fi network in college campus.

Q2. Compare 802.11 and 802.15. What are the Limitations of Bluetooth?

Definition

IEEE 802.11 → Standard for Wireless LAN (Wi-Fi).

IEEE 802.15 → Standard for Wireless Personal Area Network (WPAN), such as Bluetooth.

Comparison Between 802.11 and 802.15

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Basis	802.11	802.15
Network Type	WLAN	WPAN
Range	50-100 meters	10-30 meters
Speed	High (Mbps-Gbps)	Low to Medium
Power Consumption	High	Low
Example	Wi-Fi	Bluetooth
Coverage Area	Home/Office	Personal devices

Limitations of Bluetooth

1. Short Range (about 10 meters)
2. Low Data Rate compared to Wi-Fi
3. Limited number of devices per piconet
4. Security vulnerabilities in older versions
5. Interference in 2.4 GHz band

Diagram

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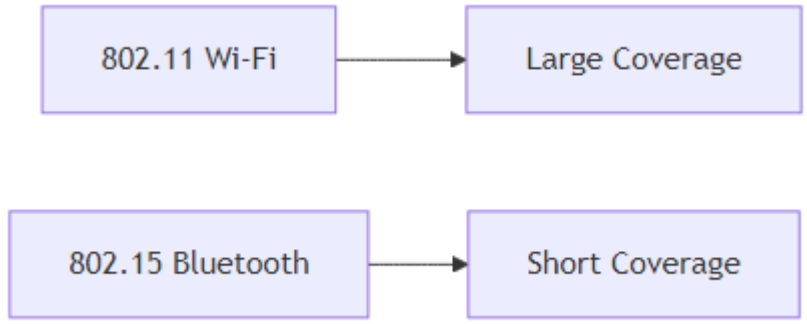

Caption: Difference in coverage area between Wi-Fi and Bluetooth.

Conclusion

802.11 is used for high-speed wireless LAN, while 802.15 is used for short-range personal communication like Bluetooth.

Q3. Explain DCF and PCF in 802.11.

Definition

DCF (Distributed Coordination Function) and PCF (Point Coordination Function) are medium access control methods in IEEE 802.11.

They control how devices access wireless medium.

1. DCF (Distributed Coordination Function)

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- Mandatory access method
- Uses CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)
- No central control
- Works in distributed manner

Working of DCF

1. Device senses channel
2. If channel free → transmit
3. If busy → wait random backoff time
4. Use ACK for confirmation

2) PCF (Point Coordination Function)

- Optional method
- Uses centralized control
- Access Point controls transmission
- Uses polling method

Working of PCF

1. Access Point sends beacon
2. AP polls stations one by one
3. Station transmits when polled

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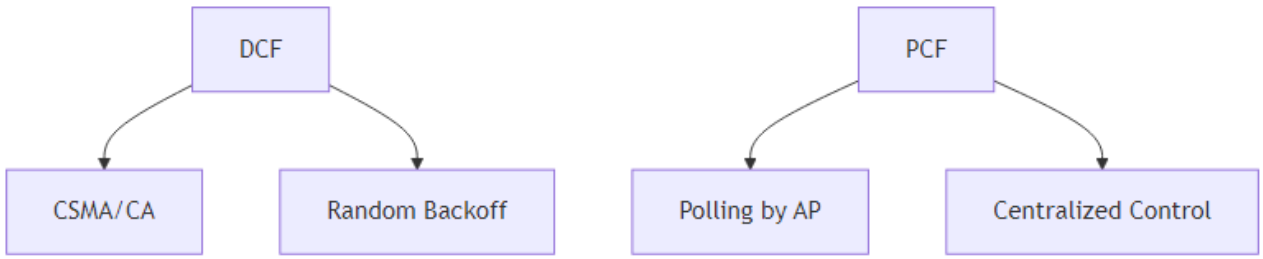
Diagram



Caption: DCF vs PCF mechanism in 802.11.

Difference Between DCF and PCF

Feature	DCF	PCF
Control	Distributed	Centralized
Mandatory	Yes	Optional
Access Method	CSMA/CA	Polling
Delay	Variable	Controlled

Conclusion

DCF uses distributed random access, while PCF uses centralized polling. DCF is commonly used in Wi-Fi networks.

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Q4. Explain Bluetooth Architecture with Diagram.

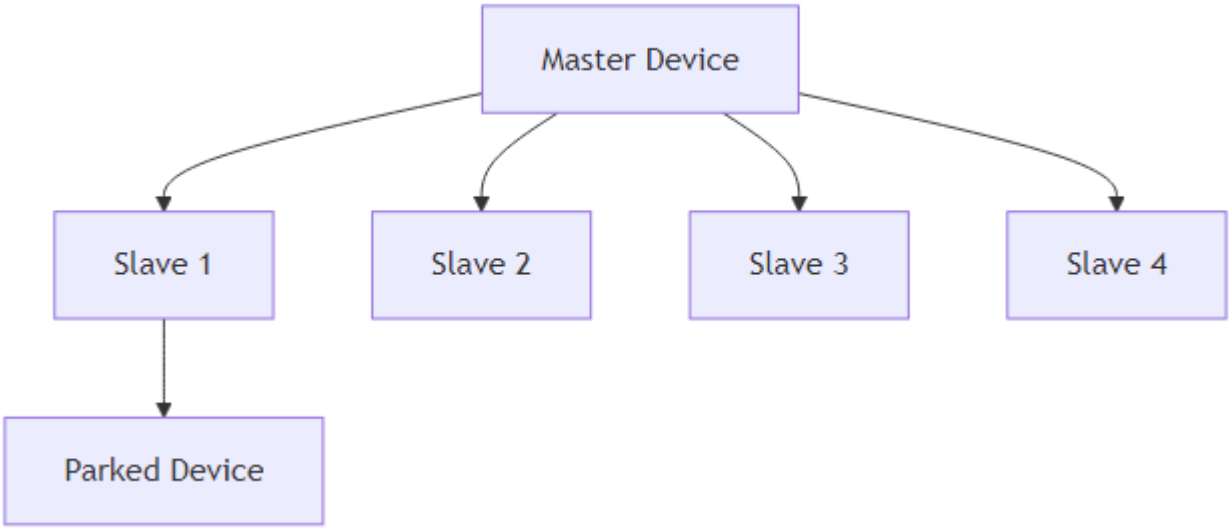
Definition

Bluetooth is a wireless communication technology based on IEEE 802.15 standard used for short-range communication between devices.

It operates in 2.4 GHz ISM band.

Bluetooth architecture is based on master-slave structure.

Diagram



Caption: Bluetooth Piconet Architecture.

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Components of Bluetooth Architecture

1. Piconet

- Basic unit of Bluetooth network
- One master and up to 7 active slaves
- Communication controlled by master

2. Scatternet

- Multiple piconets connected together
- One device can act as master in one and slave in another

3. Master Device

- Controls communication
- Provides clock and frequency hopping pattern

4. Slave Device

- Follows master instructions
- Cannot communicate directly with other slaves

5. Parked Devices

- Temporarily inactive devices

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- Can be activated later

Working Principle

- Uses Frequency Hopping Spread Spectrum (FHSS)
- Changes frequency 1600 times per second
- Provides interference resistance

Conclusion

Bluetooth architecture consists of piconets and scatternets where a master controls multiple slave devices for short-range communication.

Example

Wireless keyboard connected to laptop via Bluetooth.

Q5. What is WiMAX? List Advantages of WiMAX.

Definition

WiMAX (Worldwide Interoperability for Microwave Access) is a wireless broadband technology based on IEEE 802.16 standard.

It provides high-speed internet over long distances.

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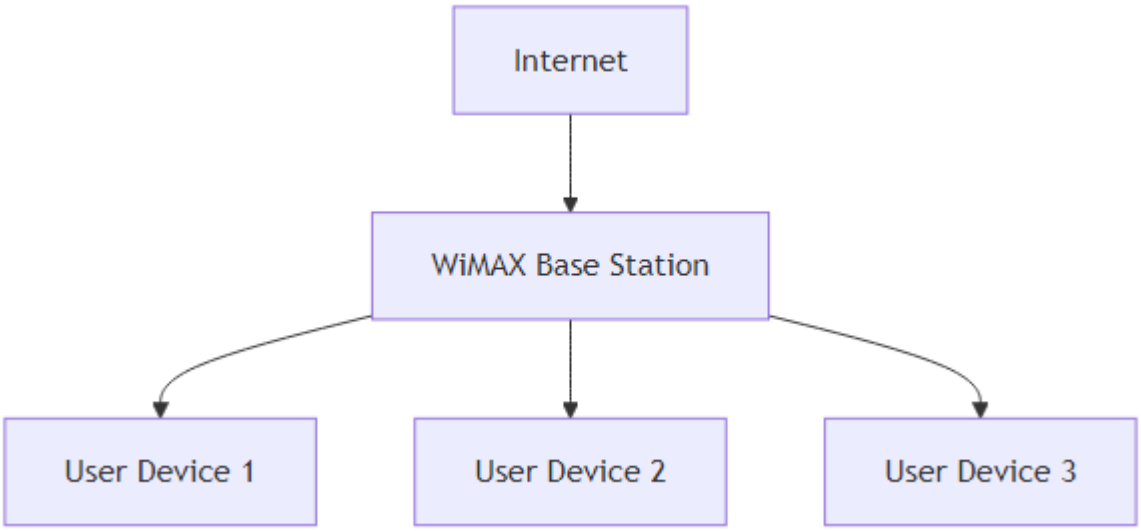
Diagram



Caption: WiMAX provides broadband connectivity over wide area.

Features of WiMAX

- Long range (up to 50 km)
- High data rates
- Supports fixed and mobile users
- Works in licensed and unlicensed bands

Advantages of WiMAX

1. Wide coverage area
2. High data transfer rate
3. Cost-effective broadband solution

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4. Supports mobility
5. Suitable for rural and remote areas
6. Better QoS support

Conclusion

WiMAX is used for metropolitan area wireless broadband with long coverage and high speed.

Example

Providing internet in rural villages without wired infrastructure.

Q6. Explain Frame Format of IEEE 802.11.

Definition

IEEE 802.11 frame format defines structure of data packet transmitted in WLAN.

Diagram

Caption: General frame structure of IEEE 802.11.

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Fields Explanation

1. Frame Control

- Type and subtype
- Control flags

2. Duration

- Time required to transmit frame

3. Address Fields (1-4)

- Source address
- Destination address
- Transmitter address
- Receiver address

4. Sequence Control

- Frame sequence number

5. Data

- Actual payload

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6. FCS (Frame Check Sequence)

- Error detection using CRC

Types of 802.11 Frames

1. Management Frame
2. Control Frame
3. Data Frame

Conclusion

IEEE 802.11 frame contains control, addressing, data, and error detection fields to ensure reliable wireless communication.

Q7. What are Hidden Station and Exposed Station Problems in WLAN?

Definition

Hidden and Exposed station problems occur in wireless networks due to limited transmission range.

1. Hidden Station Problem

Explanation

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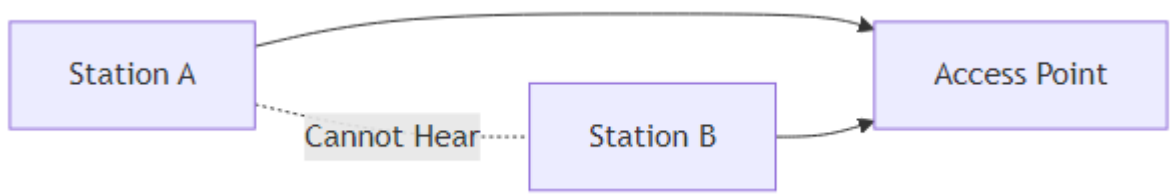
Two stations cannot hear each other but both can communicate with a common access point.

This causes collision.

Diagram



Caption: Hidden station problem causing collision.

Example

Station A and B both send data to AP at same time, causing collision.

Solution: RTS/CTS mechanism.

2. Exposed Station Problem

Explanation

One station stops transmitting because it senses another transmission, even though it would not cause collision.

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Diagram

Caption: Exposed station unnecessarily defers transmission.

Example

Station C stops transmission because it senses B transmitting, although C's transmission would not interfere.

Conclusion

Hidden station causes collision due to lack of sensing.

Exposed station reduces network efficiency due to unnecessary waiting.

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